Testing And Regression Guide

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I. Purpose

This document describes the test suite in detail and defines the regression-testing approach for future changes.

II. Test Inventory

III. Current Release Gates

The current minimum release gate is:

```
./.venv/bin/python -m unittest tests.
    test_cliquet
./.venv/bin/python -m unittest tests.
    test_sofr_curve
./.venv/bin/python -m unittest tests.
    test_smoke
```

IV. Calibration Tests

A. SOFR Curve Repricing

The curve regression test rebuilds the quoted OIS strip against the bootstrapped curve and asserts near-zero NPVs. This catches:

- accidental helper changes,
- day-count or calendar regressions,
- broken quote normalization, and
- curve interpolation changes.

B. Bermudan Helper Repricing

The Bermudan calibration test verifies that every `ql.SwaptionHelper` implied by the trade's fixed-maturity call schedule remains numerically well aligned after short-rate calibration. This is critical because a Bermudan mark can remain numerically plausible even when the calibration strip is visibly wrong. Separate schedule tests confirm that fixed-maturity Bermudans shorten the remaining underlying tenor as exercise dates roll forward, that interpolated expiries retain the correct matrix source points, and that the alternate G2++ path still builds the expected parameter set.

V. Cliquet Identity Portfolio

The cliquet regression suite does not rely on arbitrary golden numbers. Instead it uses structural identities:

1) single-period ATM call equals a vanilla call,
2) single-period ATM put equals a vanilla put,
3) single-period OTM call equals the matching vanilla call,
4) single-period ITM put equals the matching vanilla put,
5) two-period call equals the sum of two forward-start calls,
6) two-period put equals the sum of two forward-start puts,
7) four equal-quarter calls collapse to four identical quarter calls when the setup is stationary,
8) four equal-quarter puts collapse to four identical quarter puts when the setup is stationary,
9) zero-volatility call strip reduces to discounted deterministic intrinsic values, and
10) zero-volatility put strip reduces to discounted deterministic intrinsic values.

VI. Regression Strategy

A. Principles

- Prefer invariant-based tests over snapshot-style fixed numbers.
- Reprice calibration instruments directly wherever a model is calibrated.
- Keep route tests separate from core pricing tests so failures are easier to localize.
- Use deterministic seeds when Monte Carlo is involved.

B. When To Add A New Regression Test

Add a regression test when:

- a bug was fixed,
- a new instrument or market input was added,
- a pricing identity can be stated cleanly,
- a calibration path changed, or
- a UI route gained material analytical content.

VII. Performance And Runtime

The suite contains intentionally heavy tests:

- Bermudan calibration repricing is computationally expensive.
- The cliquet editor smoke test is slower than typical page renders because it runs scenario and Monte Carlo analytics.

These tests are still kept in the standard suite because the repo is a quantitative sandbox and correctness matters more than ultra-short CI runtimes.

File	Type	Coverage
`tests/test_smoke.py`	Route and integration smoke tests	Login flow, protected routes, dashboard rendering, editor rendering, model page rendering, CSV download, and market/trade update round-trips.
`tests/test_sofr_curve.py`	Quantitative calibration tests	OIS curve repricing, zero-rate and forward-strip checks, Bermudan helper repricing, workbook-reference Bermudan price checks, matrix shape checks, and risk-ladder sanity checks.
`tests/test_cliquet.py`	Structured-product identity tests	Ten cliquet reductions to simpler or deterministic cases with known answers.

VIII. Recommended Future Extensions

- add a deterministic reference test for the cliquet Monte Carlo summary path,
- add page-load budget checks if UI latency becomes a concern,
- add environment-parity smoke checks for the production WSGI entrypoint, and
- add deploy-time smoke checks against a real production health endpoint.

IX. Failure Triage

If a regression appears:

1) identify whether the failing layer is route, calibration, or payoff identity,
2) rerun the smallest targeted test file first,
3) compare normalized input state before comparing outputs,
4) verify calendar and day-count assumptions, and
5) only then inspect numerical tolerances.